

Basic Principles of Statistical Inference

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Summary

In this chapter, we discuss the fundamental principles behind two of the most frequently used statistical inference procedures: confidence interval estimation and hypothesis testing. Both procedures are constructed on the sampling distributions that we have learned in previous chapters. To better understand these inference procedures, we focus on the logic of statistical decision making and the role that experimental data play in the decision process. Numerical examples are used to illustrate the implementation of the discussed procedures. This chapter also introduces some of the most important concepts associated with confidence interval estimation and hypothesis testing, including P values, significance level, power, sample size, and two types of errors. We conclude the chapter with a brief discussion on statistical and practical significance of test results.

Key words: Hypothesis testing; P value; point and confidence interval estimation; power; sample size; significance level; simultaneous inference; student t distribution; type I and type II errors.

1. Introduction

Statistical inference is a decision-making process. Different decision-making processes follow different decision rules. Medical decisions, for example, are usually based on the physician's assessment of the patient, the physician's clinical judgment, and the physician's interpretation of treatment guidelines. A statistical decision process, or statistical inference, attempts to isolate the decision maker from his personal opinion and preference to achieve an *objective* conclusion that is supported by the data. Two commonly encountered forms of statistical inference are parameter estimation and hypothesis testing. Although each is designed to address a different type of research question, both rely upon the sample data to justify their conclusion.

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In an estimation procedure, one uses sample information to estimate the value of an unknown parameter associated with a certain population. Herein, the term *parameter* is used to describe a particular numerical characteristic of a population. For example, the mean of a population is a parameter. Denoted by μ , it depicts the central location of the population. The variance of the population, denoted by σ^2 , is another parameter. As a population parameter, σ^2 quantifies the magnitude of dispersion or the variability of the population.

The other form of statistical inference is hypothesis testing, which is primarily used to adjudicate the truthfulness of certain preconceived statements concerning the value of a population parameter. Hypothesis testing is a particularly popular form of statistical inference in biomedical research because it directly assesses the strength of data evidence either for or against a scientific proposition.

There are numerous established inference procedures. Researchers choose appropriate procedures based on the parameters of interest and experimental conditions under which the data are collected. Underlying the varying forms of statistical inference, however, are a set of principles that are common to all inference procedures. The purpose of the current chapter is to introduce these common principles and illustrate their application through several frequently used procedures, including confidence interval estimation and hypothesis testing of population means and variances. The development of these procedures is mostly based on the distributions of the sample mean and sample variance introduced in the previous chapter. Also discussed are the interpretations of inference results, errors, and power. Finally, the chapter concludes with a brief discussion on the distinction between statistical and practical significance.

2. Parameter Estimation

2.1. Point Estimation

Parameter estimation is a useful technique when the primary goal of the analysis is to estimate a certain numerical characteristic of a population. For example, a geneticist is often interested in estimating the allele frequencies of certain genes in a target population. A clinician may want to estimate the amount of viral shedding in patients with a certain infection. A biochemist may be interested in estimating the average concentration of a certain protein in a patient population. In each of these situations, the interest is to estimate a certain numerical quantity associated with a particular *population*.

With few exceptions, it is difficult for an investigator to directly assess each and every single member of the population. Therefore, directly calculating the population parameter of interest becomes logistically difficult, if not impossible. A practical way to achieve the goal of estimation is to rely on a sample of subjects, drawn from the intended population, and then use the sample data to estimate the unknown population parameter.

Although it is easy to see the appeal of such a sample-based estimation approach, there are a few important factors that might influence the quality of the resulting estimate. First among them is the representativeness of the sample. The preferred scheme is to take a *random sample* from the target population. In this chapter, all of the procedures are presented under the assumption of random sampling. Another important consideration is the optimality of the estimate, which is the main topic of the remainder of the section.

A generic description for an estimation problem is as follows: Suppose that subjects, or elements, of a certain *population* are selected through a randomized experiment, and the observations obtained from these subjects form a *sample*. Assuming that these observations follow a particular distribution with an unknown parameter, denoted as θ , we attempt to use the sample data to estimate θ . To achieve this goal, we need to have a rule for the estimation, usually expressed as a formula, that tells us how to calculate a numerical estimate based on the information provided by the sample. Such a rule, or formula, is referred to as an *estimator*. A numerical value of the estimator, usually computed from the sample, is called an *estimate*. If the true parameter is θ , we usually write an estimator (or estimate) of θ as $\hat{\theta}$. For example, an estimator for a population mean can be written as $\hat{\mu}$.

The simplest form of estimation is to calculate a single-valued point from the sample and use it as an estimator of the unknown parameter. Such an estimator is called a *point estimator*. Let's consider the following example.

Example 1

Prostate specific antigen (PSA) is a protein produced by the cells from the prostate. The blood concentration of PSA is often used as a biomarker of prostate cancer. Results under 4 ng/mL are usually considered normal. The higher the PSA level, the more likely a patient has prostate cancer. Because of this relationship, postproctectomy PSA has also been used to measure the success of the operation. **Table 1** contains the PSA levels of 30 patients who had radical proctectomy, measured 6 months after the operation.

Table 1
Prostate Specific Antigen Levels (ng/mL) in 30
Patients Who Underwent Proctectomy Measured 6
Months after Operation

0.2	0.1	0.0	0.0	0.1	0.1
0.1	0.1	0.1	0.0	0.4	0.0
0.0	0.2	0.2	0.1	2.7	0.1
0.0	0.2	1.3	0.0	0.2	0.0
0.1	0.3	0.1	0.0	0.0	0.1

If we believe that this sample of 30 patients is representative of the population of prostate cancer patients who underwent radical prostatectomy, we could use the data obtained from these patients to estimate the mean postoperation PSA value.

In a more general notation, we write $X_1, X_2, \dots, X_n$ as a random sample from a population, where X_i represents the observation from the i th experimental unit, $i = 1, 2, \dots, n$. In this example, the experimental unit is the individual patient, the sample size is $n = 30$, and sample observations are $X_1 = 0.2, X_2 = 0.1, \dots, X_{30} = 0.1$. We now use this sample to estimate the mean postoperation PSA level μ of the population of prostate cancer patients who underwent prostatectomy.

To estimate the population mean PSA level μ , we could use any of the 30 observed sample points as our estimator, that is $\hat{\mu} = X_i$ for any $i = 1, 2, \dots, 30$. Depending on which sample observation we use, the value of the estimator takes a wide range, from 0 ng/mL to 2.7 ng/mL. Alternatively, we could use the sample mean as our point estimator, $\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, because, for a given sample, $\bar{X}$ takes only a single value. For the PSA example, sample mean is $\bar{X} = 0.2267$ ng/mL. As this example demonstrates, for a given parameter, there may exist many point estimators. The question is which one of these is the “best” estimator.

In order answer this question, we must first clarify what kind of properties we desire in an estimator. Intuitively, one of the desirable characteristics that we seek in an estimator is *unbiasedness*. In other words, a good estimator should neither consistently overestimate nor underestimate the parameter. Another property that we seek in an estimator is small variability. In other words, an ideal estimator should not vary too much from sample to sample. Combining these two characteristics, we want an estimator that has small variation while maintaining its unbiasedness. This leads to the concept of minimum variance unbiased estimator (MVUE). As its name indicates, an MVUE is an unbiased estimator, and it has the smallest variance among all unbiased estimators.

In the case of population mean, it can be shown that $\hat{\mu} = \bar{X}$ is indeed an MVUE, thus is superior to the estimator based on a single sample point, $\hat{\mu} = X_i$. To understand this, let us compare $\hat{\mu} = X_i$ and $\hat{\mu} = \bar{X}$. Formally, we assume $X_1, X_2, \dots, X_n$ to be a random sample from a population with mean μ and variance σ^2 . Because X_i is a sampling point from the population, we know that the expected value and variance of X_i are μ and σ^2 , respectively; that is, $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. From the sampling distribution of the population mean, we know that $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$. This demonstrates that both single sample point (X_i) and sample mean ($\bar{X}$) are unbiased estimators of μ , but between the two, the sample mean $\bar{X}$ has a smaller variance thus should be considered as the preferred estimator. The result of this comparison is hardly surprising: the sample mean depicts the central location of the sample. If the sample truly represents the subjects in the popula-

tion, the sample mean will provide a reasonable estimate of the central location of the population. Indeed, it can be shown that $\bar{X}$ is an MVUE of $\mu(\mathbf{I})$. Therefore, as a general principle, we usually use the sample mean $\bar{X}$ to estimate the population mean μ . Extending the same principle to the case of variance, we often use sample variance $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ to estimate an unknown population variance σ^2 .

2.2. Confidence Interval Estimation

Although a good point estimator may possess desirable properties such as those just discussed, many researchers feel that without any assurance of its “correctness,” a single-valued estimate is not particularly useful in a decision-making process. A simple way to overcome this deficiency is to specify a range, rather than a single value, for the estimation of a parameter. In other words, we hope to guarantee with a high probability that the true parameter is covered by this range. Such a range is called a *confidence interval* of a parameter. The probability that guarantees the coverage is called the *confidence level*.

To define an interval mathematically, we need two values: a lower limit and an upper limit. Formally, if θ is the true parameter and $\hat{\theta}_1$ and $\hat{\theta}_2$ are the two limit values that satisfy the following criterion,

$$P(\hat{\theta}_1 \leq \theta \leq \hat{\theta}_2) = 1 - \alpha,$$

then the interval $(\hat{\theta}_1, \hat{\theta}_2)$ is said to be a $100(1 - \alpha)\%$ confidence interval estimate of θ , where P is a probability measure, $\hat{\theta}_1$ and $\hat{\theta}_2$ are called lower and upper confidence limits, respectively, and $(1 - \alpha)$ is called the confidence level. In plain English, the above equation simply says that the probability that the interval $(\hat{\theta}_1, \hat{\theta}_2)$ contains the true parameter θ is $1 - \alpha$. In practice, we often specify α as a rather small number, say $\alpha = 0.05$, thus the resulting confidence level would be high, such as 95%. Later we shall see that α represents an error rate of the inference.

For confidence intervals, we seek two properties simultaneously: The first is a high confidence level. Confidence level can be viewed as an assurance of the correctness of the estimation. Without a high confidence level, the estimation loses its credibility. The second is a narrow interval range. When the range is too wide, the estimation becomes less informative. For example, when we estimate a person to be between 5 and 95 years of age, the estimate essentially fails to convey any information on how old the person actually is. Unfortunately, these two properties are inherently contradictory: With wider interval range, we become more confident that our estimate is correct. But when the range becomes wider, our estimation becomes less useful. Therefore, a careful balance must be maintained when we derive a confidence interval.

2.2.1. Large Sample Confidence Interval for the Mean

Formally deriving confidence intervals for the population parameters exceeds the scope of the current chapter. Instead, we shall present a heuristic argument for the construction of large sample confidence interval for the population means.

Suppose we have a population with an unknown mean parameter μ . For simplicity, we assume the population variance, σ^2 , is known. To estimate the mean, we take a large random sample $X_1, \dots, X_n$, where $n \geq 30$. As described in the previous section, we use the sample mean $\bar{X} = \sum_{i=1}^n X_i / n$ as a point estimate for μ .

To find a $(1 - \alpha)100\%$ confidence interval for μ , denoted by $(\hat{\mu}_1, \hat{\mu}_2)$, we start from its definition. Suppose that $\hat{\mu}_1$ and $\hat{\mu}_2$ are two quantities that satisfy $P(\hat{\mu}_1 \leq \mu \leq \hat{\mu}_2) = 1 - \alpha$. Simultaneously subtracting $\bar{X}$ from $\hat{\mu}_1$, μ , and $\hat{\mu}_2$, and then dividing the differences by $\sigma/\sqrt{n}$, we have

$$P\left(\frac{\hat{\mu}_1 - \bar{X}}{\sigma/\sqrt{n}} \leq \frac{\mu - \bar{X}}{\sigma/\sqrt{n}} \leq \frac{\hat{\mu}_2 - \bar{X}}{\sigma/\sqrt{n}}\right) = 1 - \alpha,$$

or equivalently,

$$P\left(\frac{\bar{X} - \hat{\mu}_1}{\sigma/\sqrt{n}} \geq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{\bar{X} - \hat{\mu}_2}{\sigma/\sqrt{n}}\right) = 1 - \alpha.$$

According to the sampling distribution of $\bar{X}$, we know that when the sample size is large, $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ can be approximated by the standard normal distribution $N(0,1)$. Thus we have the following equations,

$$\frac{\bar{X} - \hat{\mu}_1}{\sigma/\sqrt{n}} = z_{\alpha/2}, \quad \text{and} \quad \frac{\bar{X} - \hat{\mu}_2}{\sigma/\sqrt{n}} = -z_{\alpha/2},$$

where $z_{\alpha/2}$ is the cutoff point corresponding with a right tail area of $\frac{\alpha}{2}$ under the standard normal distribution. Because of the symmetry of the standard normal distribution, $-z_{\alpha/2}$ is the cutoff point in the left tail.

Solving these equations, we obtain

$$\begin{aligned} \hat{\mu}_1 &= \bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \\ \hat{\mu}_2 &= \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}. \end{aligned}$$

Confidence limits calculated from the above formulas will give us the desired $100(1 - \alpha)\%$ confidence interval estimate of μ .

A few things need to be clarified before we use this confidence interval estimation procedure. The first is the assumption that σ is known. This assumption, however, is unlikely to be true because the calculation of σ depends on

μ . Because μ is unknown and is to be estimated, how could we know σ ? But for all practical purposes, this will not prevent us from using the procedure, because in large sample situations, unknown σ can be reliably approximated by the sample standard deviation s . Therefore, when the true value of σ is unknown, we calculate the large sample $100(1 - \alpha)\%$ confidence interval for the mean as

$$\bar{X} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}.$$

The second issue is the interpretation of the above confidence interval. If we use $\alpha = 0.05$, the confidence level $(1 - \alpha)$ will be 95%. In a specific experiment (or the sample generated from that experiment), the calculated confidence interval either contains μ or it does not. Although we do not know whether the confidence interval that we calculated for this particular sample contains μ or not, we do know that if we repeat the experiment many times and use the above procedure to construct confidence intervals based on the resulting samples, about 95% of the confidence intervals that we obtain will contain the true value of the population mean (2).

Example 1 (Continued)

Finally, to illustrate the use of the confidence interval procedure introduced above, we revisit the PSA example: Using the sample of 30 prostate patients, we find the point estimate of the mean postoperative PSA value to be $\bar{X} = 0.2267$. The 95% confidence interval estimate is calculated as follows:

$$\begin{aligned} & \bar{X} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}, \\ & 0.2267 \pm 1.96 \times \frac{0.5252}{\sqrt{30}}, \\ & (0.0388, 0.4146). \end{aligned}$$

Therefore, we are 95% confident that the mean postprostatectomy PSA level is between 0.0388 and 0.4146 ng/mL.

2.2.2. Student t-Distribution

As we see in the previous section, the construction of a large sample confidence interval for μ is based on the sampling distribution of $\bar{X}$. We know that when the sample size is large, the standard normal distribution $N(0,1)$ can be used to approximate the behavior of the quantity $\sqrt{n}(\mu - \bar{X})/s$. But when the sample size is not large, such approximation may not work as well. In this section, we discuss the sampling distribution of $\sqrt{n}(\mu - \bar{X})/s$ when the sample

size is small or moderate. The behavior of this distribution, called t -distribution, is very similar to that of the standard normal distribution when the sample size is large (often defined as $n \geq 30$). But it works equally well for smaller samples.

We first give a verbal description of the t -distribution. The density curve of a t -distribution is quite similar to that of the standard normal distribution. It is bell-shaped, symmetric around zero, but with tails heavier than those of the standard normal curve. The t -distribution has only one parameter, its degrees of freedom (d.f.). The larger the degrees of freedom, the smaller the variance. The mean of a t -distribution is always zero; the variance is $\text{d.f.}/(\text{d.f.} - 2)$. It can also be shown that when the degrees of freedom of the distribution approaches infinity ($\text{d.f.} \rightarrow \infty$), a t -distribution approaches the standard normal distribution.

With the t -distribution, we are now able to describe the behavior of $\sqrt{n}(\mu - \bar{X})/s$. Suppose we select a random sample $X_1, \dots, X_n$ from a normally distributed population, $N(\mu, \sigma^2)$. Let $\bar{X}$ and s^2 be the sample mean and variance, respectively. Then $T = (\bar{X} - \mu)/(s/\sqrt{n})$ follows a Student's t -distribution with $n - 1$ degrees of freedom, where n is the sample size.

In other words, *if the population is normally distributed*, statistic $T = (\bar{X} - \mu)/(s/\sqrt{n})$ has a t -distribution with $n - 1$ degrees of freedom. This is true regardless of the sample size. With this distribution, we will be able to make inferences concerning the population means, even when sample size is not large.

2.2.3. Small Sample Confidence Interval for the Mean

Mimicking the arguments in the large sample case, we construct the $100(1 - \alpha)\%$ confidence interval for μ using the t -distribution,

$$\bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}},$$

where $t_{\alpha/2}$ is the $\alpha/2$ th quantile of the t -distribution with $n - 1$ degrees of freedom.

The t confidence interval has a structure that is parallel to the large sample confidence interval based on the standard normal distribution. Although it is designed for the estimation of population means in small sample cases, it can also be used for any sample size. In fact, as the sample size increases to $n \geq 30$, the numerical value of $t_{\alpha/2}$ approaches $z_{\alpha/2}$, thus the confidence limits calculated from the small sample procedure are essentially identical to those calculated from the large sample procedure. For this reason, many standard statistical software packages only provide confidence intervals based on the t -distribution.

Example 2

B-type natriuretic peptide (BNP) is released from the cardiac ventricles in response to increased wall tension and thus can be used as a marker for congestive heart failure (3). In a local clinic, BNP levels of 20 heart failure patients were obtained from bedside assay. Fourteen of the patients were further classified as the New York Heart Association (NYHA) class II and six as class III. In the sample of NYHA class II patients, the mean BNP level was 412 pg/mL and the sample standard deviation was 231 pg/mL. For NYHA class III patients, the mean BNP level was 731 pg/mL and the sample standard deviation was 402 pg/mL. Using the sample information, we construct a 95% confidence interval for the mean BNP level of NYHA class II patients:

$$\begin{aligned} \bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}, \\ 412 \pm 2.1604 \frac{231}{\sqrt{14}}, \\ (278.6, 545.4), \end{aligned}$$

where $t_{\alpha/2} = t_{0.025} = 2.1604$ was obtained from the t -distribution with d.f. = $14 - 1 = 13$. Therefore, we conclude that based on the evidence provided by this small random sample, we are 95% confident that the mean BNP level of NYHA class II patients is between 278.6 pg/mL and 545.4 pg/mL.

2.2.4. Simultaneous Inference: Bonferroni's Multiplicity Adjustment

Similarly, one may proceed to use the information from the six NYHA class III patients to obtain a confidence interval estimate of the mean BNP level of NYHA class III patients. Although this is very easy to do computationally, one must be clear about the methodological consequence of making multiple inferences in one experiment. For one thing, the 95% confidence level that we are citing is only valid for each individual inference, not for the entire experiment. In other words, when we construct a 95% confidence interval for the mean BNP level of NYHA class II patients, we allow 5% error in our inference. When we repeat the same procedure for the NYHA class III patients, we allow another 5% error. Thus, the overall error rate for the entire experiment will be larger than the professed 5%. Intuitively, this is not difficult to understand: If there is only one inference, we have 5% chance to make a mistake. If we conduct 100 inferences, each with 5% error rate, we will make 5 mistakes in 100 inferences on average just by chance. Therefore, as the number of the inferences increases, the confidence level we have for the entire experiment decreases. This can be a very serious problem when a large number of inferences are involved. For

example, one microarray experiment may involve several thousand genes. If we set the inference error rate at 5%, we could expect to see hundreds of invalid inferences in one experiment. To guard against such inflated error rate, we often need to control the experiment-wise error (i.e., the overall chance of making an incorrect inference) rather than the error associated with a specific inference. A general class of methods for such adjustments are often referred to as simultaneous inference.

One of the popular simultaneous inference procedures is called the Bonferroni method. The basic idea of Bonferroni's approach is to reduce the individual inference error rate to a lower level so that the experiment-wise error rate can be controlled at the nominal level of $\alpha 100\%$. Specifically, if there are k inferences, we construct the $(1 - \alpha)100\%$ confidence interval using $t_{\alpha/(2k)}$ instead of $t_{\alpha/2}$.

Bonferroni's adjustment is based on the well-known Bonferroni inequality, a probability *inequality*. So the method is not exact, (i.e., the resulting experiment-wise error rate would be *no more than* α). At times, the Bonferroni adjustment may result in an experiment-wise error rate much less than the stated level. Therefore, it is often regarded as one of the more conservative multiplicity adjustment procedures.

Using the Bonferroni procedure, we can calculate the mean BNP levels for NYHA classes II and III patients as (255.6, 568.4) and (315.4, 1146.6), respectively. Note that these intervals are considerably wider than the unadjusted ones, suggesting the loss of precision as a result of the increased error rate. It should be pointed out that the Bonferroni adjustment may be too conservative (i.e., resulting intervals are too wide) for many applications. To alleviate, many alternative adjustment methods have been proposed (see **Chapter 7** for a more in-depth discussion on the subject).

2.2.5. Confidence Interval for the Variance

Occasionally, our research requires inference on population variances. In **Section 2.1**, we have stated that the point estimate of the population variance σ^2 is its sample counterpart s^2 (i.e., $\hat{\sigma}^2 = s^2$). This section describes the construction of a confidence interval estimate of σ^2 using the sampling distribution of s^2 . Specifically, let $X_1, \dots, X_n$ be a random sample from a normally distributed population with mean μ and variance σ^2 . The quantity $(n - 1)s^2/\sigma^2$ follows a chi-square distribution with $n - 1$ degrees of freedom, that is,

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{(n-1)}^2.$$

We therefore have

$$P\left(\chi_{1-\alpha/2}^2 \leq \frac{(n-1)s^2}{\sigma^2} \leq \chi_{\alpha/2}^2\right) = 1 - \alpha,$$

where $\chi_{1-\alpha/2}$, and $\chi_{\alpha/2}$ respectively correspond with the $(1 - \alpha/2)$ th and $(\alpha/2)$ th quantiles in a chi-square distribution with $n - 1$ degrees of freedom. From this, we obtain the $100(1 - \alpha)\%$ confidence interval estimate of variance σ^2 as

$$\frac{(n-1)s^2}{\chi_{\alpha/2}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}.$$

Example 3

To estimate the variation of a particular model of bone density scanner, the manufacturer of the scanners randomly selected four of its machines for testing. In measuring a standard of known density, the four selected scanners produced the following readings: 4.1, 4.0, 3.9, 3.9. Give a 95% confidence estimate of the variance of this model of bone scanner when the scanners are used to measure the standard. We first calculated the sample variance $s^2 = 0.0092$. The 95% confidence estimate of the variance of this model of bone scanner is then

$$\begin{aligned} \frac{(n-1)s^2}{\chi_{\alpha/2}^2} &\leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2} \\ \frac{(4-1) \times 0.0092}{9.3484} &\leq \sigma^2 \leq \frac{(4-1) \times 0.0092}{0.2058} \\ 0.0030 &\leq \sigma^2 \leq 0.1341, \end{aligned}$$

where 0.2058 and 9.3484 are the 2.5th and 97.5th percentiles, respectively, of the chi-square distribution with 3 degrees of freedom. Based on the above calculation, we can claim with 95% confidence that the variance of the measuring instruments is between 0.003 and 0.1341.

2.2.6. One-Sided Confidence Intervals

The discussion of confidence intervals so far has been of two-sided intervals. That is, we provide lower and upper limits on the parameter of interest. We can also define one-sided confidence intervals that consist only of a lower confidence limit or only an upper confidence limit. A lower one-sided $100(1 - \alpha)\%$ confidence interval estimate of θ satisfies the following criterion

$$P(\hat{\theta}_1 \leq \theta) = 1 - \alpha.$$

An upper one-sided $100(1 - \alpha)\%$ confidence interval estimate of θ satisfies the following criterion

$$P(\theta \leq \hat{\theta}_2) = 1 - \alpha.$$

One-sided confidence intervals are calculated using the similar formulae as for two-sided intervals. Instead of using the $\alpha/2$ and $1 - \alpha/2$ quantiles of the reference distributions, we use either the α or the $1 - \alpha$ quantile.

Example 2 (Continued)

The lower confidence limit of a one-sided 95% confidence interval for the mean BNP level of NYHA class II patients is

$$\begin{aligned} & \bar{X} - t_\alpha \frac{s}{\sqrt{n}}, \\ & 412 - 1.7709 \frac{231}{\sqrt{14}}, \\ & (302.7, \infty), \end{aligned}$$

where $t_\alpha = t_{0.05} = 1.7709$ was obtained from the t -distribution with d.f. = $14 - 1 = 13$. We would conclude that we are 95% confident that the mean BNP level of NYHA class II patients is greater than 302.7 pg/mL. The upper confidence limit of a one-sided 95% confidence interval for the mean BNP level of NYHA class II patients is

$$\begin{aligned} & \bar{X} + t_\alpha \frac{s}{\sqrt{n}}, \\ & 412 + 1.7709 \frac{231}{\sqrt{14}}, \\ & (-\infty, 521.3). \end{aligned}$$

We would conclude with 95% confidence that the mean BNP level of NYHA class II patients is less than 521.3 pg/mL. In practice, we would use one of the one-sided confidence intervals or a two-sided interval.

3. Hypothesis Testing

3.1. Understanding Hypothesis Testing

Another form of statistical inference is hypothesis testing. Hypothesis testing uses sample information to decide the truthfulness of a prespecified statement concerning a certain population parameter. The procedure leads to a decision of either rejecting or not rejecting the statement. The statement being tested is often referred to as a *hypothesis*.

A typical hypothesis testing procedure involves 5 steps:

1. Formulating the null hypothesis (denoted as H_0) and the alternative hypothesis (H_a).
2. Specifying the significance level (α).
3. Computing the value of the test statistic.
4. Determining the rejection region.
5. Stating a conclusion.

The essence of this 5-step procedure is to specify a proposition and then use the sample data to disprove it. The first step is to formulate the proposition of interest into a testable hypothesis. To be more exact, we formulate two contradicting statements: the *null hypothesis* (H_0) and the *alternative hypothesis* (H_a). By giving two contradicting hypotheses, we will force the decision maker to either reject or not reject the null hypothesis. Theoretically speaking, it should not matter which statement is specified as the “null” and which as the “alternative.” However, for the convenience of defining the types of errors later in the chapter, we consider statements representing the *status quo*, *no change*, or *equal* as the null. The alternative is then reserved for statements implying a *change*, *inequality*, *greater than*, *less than*, and so forth. Under this convention, when we formulate the hypotheses into mathematical expressions, the equals sign “=” always appears in the null statement.

Using a generic notation, we denote the parameter of interest as θ . For tests concerning θ , there are three commonly encountered pairs of hypotheses:

1. two-sided test: $H_0: \theta = \theta_0$ versus $H_a: \theta \neq \theta_0$;
2. one-sided test: $H_0: \theta \geq \theta_0$ versus $H_a: \theta < \theta_0$;
3. one-sided test: $H_0: \theta \leq \theta_0$ versus $H_a: \theta > \theta_0$,

where θ_0 is the hypothesized value for the test.

In theory and in practice, hypotheses of interest come in various forms and are by no means restricted to the three pairs listed above. In this chapter, however, we shall use the hypotheses of such simple forms to illustrate a typical testing procedure. The same reasoning process is easily extended to hypotheses of difference forms.

The second step in testing a hypothesis is to specify a *significance level*. The significance level (denoted as α) is a *prespecified* maximum probability of incorrectly rejecting the null hypothesis when it is true. It represents the maximum amount of risk that one is willing to take when he or she rejects the null hypothesis. Naturally, we would like to set this level low. For example, the most commonly used significance level is $\alpha = 0.05$. As we will discuss later in the chapter, the significance level represents the maximum allowable type I error of the inference procedure.

The third step is to compute the value of the test statistic. Hypothesis testing is about making a decision using the information that comes from a sample. A

test statistic is a quantity that measures the discrepancy between the null hypothesis and the sample data. Its value depends on the sample¹ and its sampling distribution has to be known when the null hypothesis is true. In interpreting the statistic, a “large” value of the test statistic implies a more pronounced disagreement between the data information and the null hypothesis, suggesting that the data do not support the null statement and the null statement should be rejected.

But how large is large? That is the key question we address in the fourth step. From an operational point of view, we need a rule to help us to decide whether the test statistic is indeed too large for the null hypothesis to be true. Such a rule is often expressed as a *rejection region*. Loosely speaking, a rejection region is simply a range of values such that we reject the null hypothesis if the test statistic falls in this region.

Finally in the fifth step, we state our conclusion of the test in the context of the problem.

The basic idea behind these 5 steps is very similar to that of a *proof by contradiction*. Let’s compare the two:

1. Proof by contradiction: We first assume that **A** is true. From **A** we derive **B**. If **B** is known to be false, then we claim that **A** *cannot* be true. Thus the assumption that **A** is true must be rejected.
2. Hypothesis testing: We first assume that the null hypothesis is true. Under this assumption, the test statistic follows a known sampling distribution. If the test statistic takes a value that implies the data are *probably* contradicting the null hypothesis, then we reject the null hypothesis and conclude that H_0 is *unlikely* to be true.

To illustrate, we consider a classic example of proof by contradiction by Euclid (c.325–265 BC):

Example 4

Suppose that we want to prove that there does not exist a “largest prime number.” We first assume that there exists a largest prime number, which we refer to as p . Then we let x be 1 plus the product of all prime numbers between 1 and p (i.e., $x = 1 \cdot 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdots p + 1$). Clearly, x has no prime factors between 1 and p because dividing x by any of the prime factors would leave a remainder of 1. Hence by definition, x is a prime number. Because $x > p$, we have found a prime number that is larger than the “largest prime number” p . Therefore, p cannot be the largest prime number, and we must reject the assumption that there exists a largest prime number.

¹ So it is a *statistic* and a *random* variable!

The logic of a hypothesis testing is almost entirely parallel to that used in the proof of contradiction. The only difference is that in hypothesis testing, we use probability statements instead of the “absolute” judgments.

3.2. One-Sample t-Test

We now reconsider the PSA example as a testing problem. In the process, we will construct a test statistic for inference about the population mean.

As we discussed earlier, prostate specific antigen (PSA) is an important biomarker for prostate cancer. Ideally, the postoperation PSA value should drop to an undetectable level (i.e., 0 ng/mL) shortly after surgery. But in most patients, PSA values will gradually increase over time. Visually examining the PSA values in the sample, we see that 1 year after surgery, the PSA values in most patients are no longer zero. The question is whether the mean PSA level of this patient population is significantly higher than zero. To answer this question, we test the following hypotheses: $H_0: \mu = 0$ ng/mL versus $H_a: \mu > 0$ ng/mL, where μ is the mean PSA level measured 12 months after the surgery. The rejection of the null hypothesis implies that the PSA has become detectable.

In a more generic notation, the null hypothesis can be written as $H_0: \mu = \mu_0$ and the alternative as $H_a: \mu > \mu_0$. We assume that the random sample $X_1, \dots, X_n$ comes from a normal population. When the null hypothesis is true, the population mean is $\mu = \mu_0$ and the quantity $T = \sqrt{n}(\bar{X} - \mu_0)/s$ follows a t -distribution with $n - 1$ degrees of freedom.

Let us assume that H_0 is true. Because the density of a t -distribution can be characterized as a bell-shaped curve that is symmetrical around zero under the null hypothesis, $T = \sqrt{n}(\bar{X} - \mu_0)/s$ is likely to take a value near the distribution center (zero). Therefore, in the unlikely event of an extremely large T value, we have to wonder whether we have made a mistake in assuming $H_0: \mu = 0$ to be true. In fact, the further away T is from zero, the more likely that H_0 is false. Following this logic, when T takes a very large value, we will have no other choice but to reject the null hypothesis H_0 .

Examining the structure of T , we see that the magnitude of the T value reflects the level of discrepancy between the null hypothesis $H_0: \mu = \mu_0$ and the sample information. When the two agree, the T value tends to be small in the absolute value. When the two disagree, the magnitude of $|T|$ will be large (i.e., T takes on an unlikely value). When T takes an unlikely value, we reject H_0 . It is now clear that T plays a role that is instrumental in deciding whether we should reject the null hypothesis or not. In statistics, quantities such as T are often referred to as *test statistics*. They are statistics because their values depend on the sample data. They can be used in hypothesis testing because they have known sampling distributions under the null hypotheses so we know which values are likely and which ones are unlikely.

Table 2
Reject Regions for the Test of Population Mean μ

Alternative hypothesis	Rejection region
$\mu < \mu_0$	$T < -T_\alpha$
$\mu > \mu_0$	$T > T_\alpha$
$\mu \neq \mu_0$	$T < -T_{\alpha/2}$ or $T > T_{\alpha/2}$

With the value of the test statistic T calculated, we are ready to decide on the hypotheses. The range of T values that leads to the *rejection of the null hypothesis* is called the *rejection region*. Intuitively, the reject regions have to be in the tails of the t -curve. **Table 2** summarizes the rules we use to define the rejection regions.

Now we complete the PSA example following the 5-step procedure. Let μ be the mean PSA level measured 1 year after prostatectomy.

1. $H_0: \mu = 0 \text{ ng/mL}$ versus $H_a: \mu > 0 \text{ ng/mL}$.
2. We set the significance level at $\alpha = 0.05$.
3. Computing the value of the test statistic, we have

$$T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{0.2267 - 0}{0.5252/\sqrt{30}} = 2.3642.$$

4. For the t -distribution with d.f. = $30 - 1 = 29$ degrees of freedom, $T_{0.05} = 1.699$. So we should reject the null hypothesis if $T > 1.699$. Because $T = 2.3642 > 1.699$, we reject the null hypothesis.
5. The sample evidence indicates that 1 year after prostatectomy, the mean PSA level in prostate cancer patients is significantly greater than zero.

3.3. An Alternative Decision Rule: P Value

The 5-step procedure introduced above involves the use of a *preselected* significance level α . So the maximum tolerable risk of incorrectly rejecting the null hypothesis is *subjectively* determined. The procedure is not particularly flexible, because in rejecting the null hypothesis, we only know that the error rate is not more than α , and have no idea how large the error rate actually is.

In hypothesis testing, the actual probability of incorrectly rejecting the null hypothesis given the sample observations is often called the P value or the *observed* significance level of the test. By definition, if a test yields a “large” P value, then there will be a “large” risk of committing an error in rejecting the true null hypothesis. A small P value, on the other hand, implies that there is little chance of making a mistake if the null hypothesis is rejected. Therefore,

as a decision rule, *we reject the null hypothesis when the P value is small*. When this happens, we say the test result is statistically significant.

The computation of the P value in a hypothesis test is simple: Because we reject the null hypothesis only when the value of the test statistic is in the tail(s), the tail area defined by the test statistic represents the probability of incorrectly rejecting the null hypothesis. To be more exact, we only need to compute the corresponding tail area under the t density curve using the value of the test statistic T as a cutoff point in a one-tail situation. For the cases involving two-sided tests, we simply multiply the one-side tail area by 2 to take into account the areas in both tails.

Thus, we have an alternative procedure of hypothesis testing using the P value:

1. Formulating the null hypothesis (H_0) and the alternative hypothesis (H_a).
2. Computing the value of the test statistic.
3. Computing the P value.
4. Stating a conclusion.

We now illustrate this new procedure by reanalyzing the PSA data. Let μ be the mean postoperation PSA level of prostate cancer patients.

1. $H_0: \mu = 0 \text{ ng/mL}$ versus $H_a: \mu > 0 \text{ ng/mL}$.
2. Computing the value of the test statistic, we have

$$T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{0.2267 - 0}{0.5252/\sqrt{30}} = 2.3642.$$

3. The tail area that corresponds with $T \geq 2.3642$ is $P(T \geq 2.3642) = 0.0125$. This P value of 0.0125 says if we reject the null hypothesis, there will be a 1.25% chance for us to make a mistake. Because the risk for an error is very small, we choose to reject the null hypothesis.
4. This testing procedure again confirms that the mean PSA level in prostate cancer patients is greater than zero 1 year after prostatectomy.

In practice, data analysts usually have the aid of various statistical computing programs to carry out computational tasks required by inference procedures. Thus, the issue is often not so much about the computation of the P values but how to interpret them. For example, in the PSA data example, we have a P value of 0.0125 and we decide that 1.25% error rate is a small risk to take when we reject the null hypothesis. But there is no scientifically compelling reason for others to share our decision rule. In fact, a frequently asked question is when is a P value considered small and, therefore, when should a significant test result be declared? Unfortunately, there is no short answer to this question that is universally agreed upon by all scientists. Whereas some scientists favor the adoption of 0.05 as the threshold for the rejection of H_0 in all situations, others argue

for more flexibility. From the perspective of statistical decision making, the P value is simply an indicator of the strength of data evidence against the null hypothesis. The stronger the evidence against H_0 , the smaller the P value and the less the chance of incorrect rejection. Under this general principle, with a given P value, the investigator always retains the flexibility to reject or not to reject the null hypothesis. But in case she decides to reject, we will know the chance of a mistake. By reporting this P value, the investigator has given the readers a chance to assess the strength of the evidence. This is why many scientists consider the P value method more advantageous than the rejection region method.

3.4. Errors, Power, and Sample Size

No decision-making process is completely error-free. Although our discussion has so far focused on the error rate of incorrectly rejecting a null hypothesis, statistical testing is actually subject to more than one type of error. One makes a mistake in rejecting the null hypothesis if the null is true (called a type I error); and one makes a different kind of mistake by not rejecting the null when it is false (called a type II error).

Table 3 summarizes the situations leading to the occurrence of errors.

Here, α is the type I error rate (i.e., the conditional probability of rejecting the null hypothesis given that the null hypothesis is true); β is the type II error rate (i.e., the conditional probability of failure to reject the null hypothesis when the null is false). Ideally, one hopes to keep both α and β low. Journal editors routinely require the authors to specify the maximum value of the type I error rate (α) or the P values for statistical tests, which is not particularly difficult to do. But they rarely ask for a type II error rate β . Indeed, specifying a type II error rate is a significantly more difficult task to accomplish, because this error can happen under many different values of the alternative hypothesis. Once the data are collected, we are left with no real options to control this error. Therefore, we often try to control it in the design stage of the experiment. To be more exact, we can control the *power* ($1 - \beta$) of the test by properly adjusting the sample size n . We usually require the power to be at least 80% (which implies that the type II error rate β is at most 0.2).

Table 3
Types of Errors in Hypothesis Tests

Decision	H_0 is true	H_a is true
Reject H_0	Type I error (α)	Correct decision ($1 - \beta$)
Do not reject H_0	Correct decision ($1 - \alpha$)	Type II error (β)

For a large sample test of a population mean μ , the sample size required to guarantee a $(1 - \beta)$ power at significance level α is

$$n = \left[\frac{(z_{\alpha/2} + z_{\beta})s}{\mu_1 - \mu_0} \right]^2 = \left[\frac{(z_{\alpha/2} + z_{\beta})s}{\Delta} \right]^2,$$

where Δ is the difference to be detected by the test, and $z_{\alpha/2}$ and z_{β} are the z values corresponding with the right tail areas of $\alpha/2$ and β , respectively, in a standard normal distribution.

Again, we consider the PSA example.

Example 5

Assuming a standard deviation of 0.6 ng/mL, if we intend to detect a 0.20 pg/mL change in the mean PSA level with 0.05 significance level and 80% power in a two-sided test, how many patients do we need for the test?

Sample size calculation for the PSA example: Following the above formula, the required sample size can be calculated as

$$n = \left[\frac{(z_{\alpha/2} + z_{\beta})s}{\Delta} \right]^2 = \left[\frac{(1.96 + 0.84) \times 0.6}{0.2} \right]^2 \approx 71.$$

3.5. Statistical Significance and Practical Significance

When we reject a null hypothesis with a small P value, we say the result is *statistically significant*. In most practical situations, this simply means that a change (or difference) has been detected by our test. It does not imply that the change (or the difference) itself is of any practical importance. For example, it is possible that we detect a change of 2 mmHg in systolic blood pressure between two patient groups. The result is statistically significant, but clinically the change is quite trivial.

On the other hand, when there is a change of real clinical significance, a statistical test may not be able to detect it if the sample size is not large enough (insufficient sample size leads to the lack of testing power)! Keep in mind that these two concepts originate from two different decision-making processes, and we should not equate one with the other.

Having studied principles of testing a statistical hypothesis, we now face the ultimate challenge: how to make our statistical tests practically relevant? This question should be addressed from two different aspects:

1. The hypotheses should reflect practically significant changes. In other words, the amount of change we intend to test should be of some practical importance.
2. When we design the experiments, we should always choose a sample size that gives us a reasonable chance to detect a practically meaningful change. In the

absence of a careful sample size estimate, a statistically nonsignificant test result should never be interpreted as evidence of “no difference.”

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Title: Standard Error and Estimation

Introduction to Standard Error:

- Standard error is a measure of the precision or variability of an estimator (e.g., sample mean, sample proportion) in estimating a population parameter.
- It quantifies the average amount of uncertainty or sampling variation associated with the estimator – it is the standard deviation of a statistic.

Calculation of Standard Error:

The standard error is typically calculated using the standard deviation of the sample statistic and the sample size.

- For the sample mean, the standard error (SE) is given by $SE = (\text{standard deviation} / \sqrt{n})$, where n is the sample size.
- For other estimators, such as proportions or regression coefficients, the formula for standard error may differ but follows a similar concept.

Interpretation of Standard Error:

- A smaller standard error indicates less variability or more precision in the estimator, suggesting a more reliable estimate of the population parameter.
- A larger standard error indicates more variability or less precision in the estimator, implying increased uncertainty in estimating the population parameter.

Estimation using Standard Error:

Standard error is crucial for constructing confidence intervals and conducting hypothesis tests.

- **Confidence Intervals:** Standard error is used to calculate the margin of error in constructing confidence intervals. For example, a 95% confidence interval for the population mean is typically estimated as $(\text{sample mean} \pm 1.96 * SE)$, where 1.96 represents the critical value from the standard normal distribution for a 95% confidence level.
- **Hypothesis Testing:** Standard error helps in determining the test statistic and calculating the p-value. It is utilized to assess whether the observed difference between sample statistics and population parameters is statistically significant.

Examples from Biological Scenarios:

Example 1: Estimating Mean Body Weight:

Researchers collect a random sample of body weights (in kilograms) from a population of animals. The sample mean is 15 kg, and the sample standard deviation is 2 kg. The standard error of the sample mean can be calculated as $SE = (2 / \sqrt{n})$, where n is the sample size.

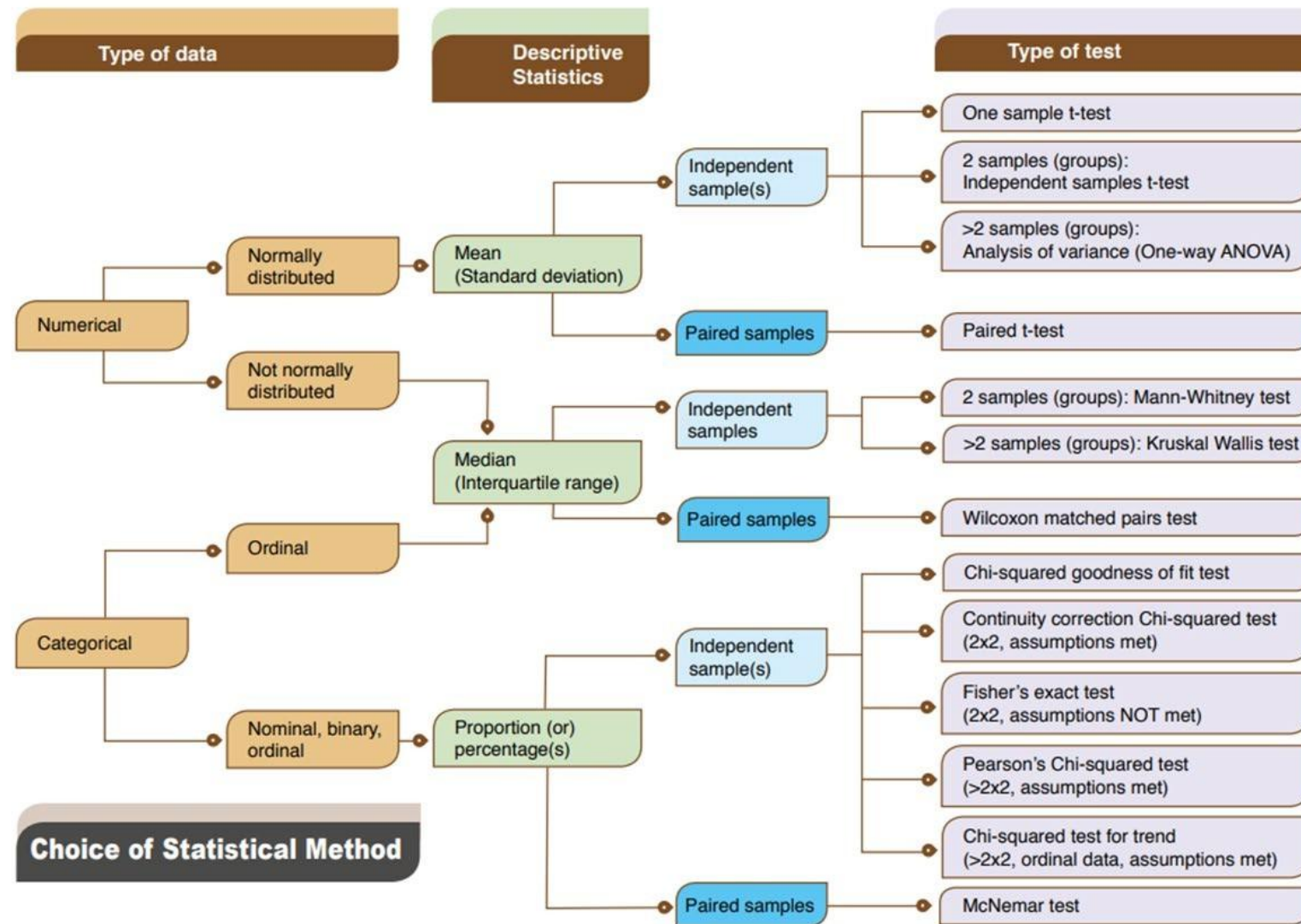
With a larger sample size, the standard error decreases, indicating a more precise estimate of the population mean. A smaller standard error suggests that the sample mean is likely to be close to the population mean.

Example 2: Estimating Proportion of Disease:

In a study investigating disease prevalence in a population, a random sample of individuals is selected, and the proportion of individuals with the disease is determined. The sample proportion is 0.25, and the sample size is 200. The standard error of the sample proportion can be calculated using the formula $SE = \sqrt{[(p(1-p)) / n]}$, where p is the sample proportion and n is the sample size. The standard error reflects the precision of the estimated disease prevalence. A smaller standard error implies a more precise estimate of the population proportion, indicating a narrower range of possible values.

Understanding standard error is essential in assessing the reliability and precision of estimators in biological scenarios. It enables researchers to construct meaningful confidence intervals, make accurate inferences, and determine the statistical significance of findings. By accounting for sampling variation, standard error provides valuable insights into the uncertainty associated with estimators and enhances the validity of statistical analyses in biological research.

Choice of Statistical Test



Module 6: Estimation & Confidence Intervals

Module Assignment

Source notes cover parameter estimation, point estimates, confidence intervals, standard error and interpretation of uncertainty.

1. Using a clinical dataset, estimate a population mean and a population proportion.
2. Calculate or obtain a 95% confidence interval for each estimate.
3. Repeat the interpretation for a 90% and 99% confidence level where software/calculator support is available.
4. Explain, in clinical terms, what the width of each confidence interval says about precision.
5. Distinguish a point estimate from a confidence interval.

Submission guidance: Show working where calculations are required. Interpret statistical results in clinical language. Use Excel, Jamovi or SPSS where appropriate.